Fourier-level lower bound for every low-degree uniform approximator on fractional subcubes

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doi:10.4230/LIPIcs.CCC.2021.10; CC BY 4.0

Editorial note. Editable transcription of the official CC BY author approximation-converse result and full bearing Sections2.1/3/3.1 context. Independent public copy blanks all unselected material while retaining native original locator lines; formal chapter, statement and equation offsets restored. Author mathematical statements and proofs unchanged. Standard Chebyshev extremal theorem cited rather than recursively reproved. All production and validation solely on Yupeng.

Allowed standard prerequisites (editorial).

- Boolean Fourier analysis, univariate Taylor theorem, standard bounded-wise independence/small-bias generators and prior polarizing-walk amplification. Complete originalSection2 definitions and cited standard amplificationTheorem14 included; all source-local Fourier approximation and random-restriction proofs retained

2 Preliminaries

As in [CHHL19] and [CHLT19], we study PRGs for classes $\mathcal{F}$ of n-variate Boolean functions that are closed under restriction (that is, fixing any subset of the input variables of a function in the class yields a function that remains in the class).

2.1 Fourier Analysis

We briefly recall basic Fourier analysis: any Boolean function $f : \{-1,1\}^n \to \{-1,1\}$ admits a unique multilinear expansion, also known as the *Fourier expansion*, given by

$$f(\mathbf{x}) = \sum_{S \subseteq [n]} \hat{f}(S)\mathbf{x}^S, \tag{2}$$

where we write $\mathbf{x}^S \triangleq \prod_{i \in S} x_i$. The Fourier coefficient $\hat{f}(S)$ is given by

$$\hat{f}(S) = \mathbb{E}_{\mathbf{X} \sim \{-1,1\}^n}[f(\mathbf{X})\mathbf{X}^S].$$

For more on Fourier analysis of Boolean functions, see the excellent book by O'Donnell [O'D14]. One may thus extend the domain of f to $[-1,1]^n$, where

$f(\mathbf{x})$ for arbitrary $\mathbf{x}$ is evaluated according to the expression in (2). Note that in this case, $f(\mathbf{0}) = \hat{f}(\emptyset) = \mathbb{E}_{\mathbf{U}_n}[f(\mathbf{U}_n)]$. One of the main parameters of interest from the Fourier expansion for this framework is the following:

Definition 7. The *level-k mass of a Boolean function f* is

$$L_{1,k}(f) \triangleq \sum_{S \subseteq [n]:|S|=k} |\hat{f}(S)|,$$

and the *level-k mass of a class $\mathcal{F}$* is $L_{1,k}(\mathcal{F}) \triangleq \max_{f \in \mathcal{F}} L_{1,k}(f)$.

In this work, we will show how to construct PRGs whose seed length depends on the following, smaller quantity:

Definition 8. For any multilinear polynomial $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(\mathbf{x}) = \sum_{S \subseteq [n]} \hat{f}(S)\mathbf{x}^S$, define the level-$k$ part by

$$f_k(\mathbf{x}) \triangleq \sum_{S \subseteq [n]:|S|=k} \hat{f}(S)\mathbf{x}^S,$$

and further define $f_{<k}(\mathbf{x}) \triangleq \sum_{i=0}^{k-1} f_i(\mathbf{x})$ and $f_{\geq k}(\mathbf{x}) \triangleq \sum_{i=k}^{n} f_i(\mathbf{x})$. Then we define the *level-k absolute Fourier sum of f* by

$$M_k(f) \triangleq \max_{\mathbf{x} \in [-1,1]^n} \left| \sum_{S:|S|=k} \hat{f}(S)\mathbf{x}^S \right| = \max_{\mathbf{x} \in \{-1,1\}^n} \left| \sum_{S:|S|=k} \hat{f}(S)\mathbf{x}^S \right|$$

and analogously define $M_k(\mathcal{F}) \triangleq \max_{f \in \mathcal{F}} M_k(f)$ for a class $\mathcal{F}$.

Note that the equality arises by multilinearity, and clearly we have $M_k(f) \leq L_{1,k}(f)$ by the triangle inequality. Without loss of generality, we may further assume that our class is closed under flipping the image, i.e. we may suppose that $f \in \mathcal{F}$ if and only if $-f \in \mathcal{F}$; this transformation does not change either $L_{1,k}(f)$ or $M_k(f)$, and therefore the same bound on the class still holds when completing it to include all such functions. If this is the case, we get the more striking identity:

Lemma 9. *Suppose that $\mathcal{F}$ is closed under negation of variables and that $f \in \mathcal{F}$ implies $-f \in \mathcal{F}$. Then*

$$M_k(\mathcal{F}) = \max_{f \in \mathcal{F}} \sum_{S:|S|=k} \hat{f}(S) = \max_{f \in \mathcal{F}} f_k(\mathbf{1}).$$

To see why this holds, simply note that if $(f, \mathbf{z}) \in \mathcal{F} \times \{-1,1\}^n$ is a maximizer in the definition of $M_k(\mathcal{F})$ (where we may now assume that the sign is positive), then by replacing the function $f(\mathbf{x})$ with $g(\mathbf{x}) = f(\mathbf{x} \circ \mathbf{z})$, where $\circ$ denotes componentwise multiplication, we have

$$M_k(\mathcal{F}) = \left| \sum_{S:|S|=k} \hat{f}(S)\mathbf{z}^S \right| = \sum_{S:|S|=k} \hat{g}(S) = \max_{h \in \mathcal{F}} \sum_{S:|S|=k} \hat{h}(S).$$

In particular, it suffices to bound the *unsigned level-k Fourier sum* of such a class.

Lastly, we require the following notion:

Definition 10. Let $\mathcal{F}$ be a class of n-variate multilinear polynomials that is closed under restrictions. Define $\mathrm{conv}(\mathcal{F})$ as the convex closure of $\mathcal{F}$,

$$\mathrm{conv}(\mathcal{F}) \triangleq \left\{ \sum_{f\in\mathcal{F}} \lambda_f f \;\middle|\; \sum_{f\in\mathcal{F}} \lambda_f = 1, \lambda_f \geq 0 \ \forall f \in \mathcal{F} \right\}.$$

We briefly note the following two elementary facts: first, by the assumption that $\mathcal{F}$ is closed under restrictions, the same is true of $\mathrm{conv}(\mathcal{F})$. The second is the following simple claim:

Lemma 11. *For any class $\mathcal{F}$ of Boolean functions, $M_k(\mathcal{F}) = M_k(\mathrm{conv}(\mathcal{F}))$.*

Proof. One direction is obvious: as $\mathcal{F} \subseteq \mathrm{conv}\mathcal{F}$, clearly $M_k(\mathcal{F}) \leq M_k(\mathrm{conv}(\mathcal{F}))$. In the other direction, let $g = \sum_{f\in\mathcal{F}} \lambda_f f$ be an arbitrary element of $\mathrm{conv}(\mathcal{F})$, where $\lambda_f \geq 0$ and $\sum_{f\in\mathcal{F}} \lambda_f = 1$. Then

$$\begin{aligned}
M_k(g) &= \max_{\mathbf{x}\in\{-1,1\}^n} \left| \sum_{S\subseteq[n]:|S|=k} \widehat{g}(S)\mathbf{x}^S \right| \\
&= \max_{\mathbf{x}\in\{-1,1\}^n} \left| \sum_{S\subseteq[n]:|S|=k} \Big(\sum_{f\in\mathcal{F}} \lambda_f \widehat{f}(S) \Big) \mathbf{x}^S \right| \\
&\leq \sum_{f\in\mathcal{F}} \lambda_f \max_{\mathbf{x}\in\{-1,1\}^n} \left| \sum_{S\subseteq[n]:|S|=k} \widehat{f}(S)\mathbf{x}^S \right| \\
&\leq \max_{f\in\mathcal{F}} M_k(f).
\end{aligned}$$

The reverse inequality immediately follows. □

3 Low-Degree Polynomial Approximations on Subcubes

Throughout this section, we assume that $\mathcal{F}$ is a class of n-variate Boolean functions closed under restrictions. As mentioned above, the main result from which we derive our improvements in constructing pseudorandom generators is essentially a statement about low-degree polynomial approximations on subcubes $[-c,c]^n$ for $c < 1$. We remark that this setting is equivalent to approximating *noisy* versions $T_c f$ on $[-1,1]^n$, where T_ρ is the ρ-noise operator. This is because for any $\mathbf{y} \in [-c,c]^n$, we can write $\mathbf{y} = c\mathbf{x}$ for some $\mathbf{x} \in [-1,1]^n$ and

$$f(\mathbf{y}) = f(c\mathbf{x}) = \sum_{S\subseteq[n]} \hat{f}(S)(c\mathbf{x})^S = \sum_{S\subseteq[n]} c^{|S|}\hat{f}(S)\mathbf{x}^S = T_c f(\mathbf{x}).$$

In general, given any $k \leq n$, $c \geq 0$, and any $f \in \mathcal{F}$, let $\varepsilon_{c,k}(f)$ be defined by

$$\varepsilon_{c,k}(f) \triangleq \inf_{g:\deg(g)<k} \max_{\mathbf{x}\in[-c,c]^n} |f(\mathbf{x}) - g(\mathbf{x})|, \tag{3}$$

and extend the definition to function classes by

$$\varepsilon_{c,k}(\mathcal{F}) \triangleq \max_{f\in\mathcal{F}} \varepsilon_{c,k}(f).$$

Now, given $\varepsilon > 0$, $k \leq n$, and the class $\mathcal{F}$, define $c_k(\mathcal{F}, \varepsilon)$ by

$$c_k(\varepsilon, \mathcal{F}) \triangleq \max\{c \geq 0 : \varepsilon_{c,k}(\mathcal{F}) \leq \varepsilon\}.$$

In words, $c_k(\varepsilon, \mathcal{F})$ measures how small a hypercube we must take to ensure that for every function in our class, there exists a degree-$(k-1)$ approximating polynomial that agrees with f up to a uniform ε error on the subcube $[-c, c]^n$; by multilinearity, it actually suffices that this holds at the extreme points $\{-c, c\}^n$. Note that (3) can be formulated as a linear program and its optimal solution is the best low-degree ℓ_∞-approximation to f.

The main technical claim in this section is that we bound $c_k(\varepsilon, \mathcal{F})$ in terms of $M_k(\mathcal{F})$. Specifically, we show that for any class $\mathcal{F}$ that is closed under restrictions, truncating the Fourier expansion of a function $f \in \mathcal{F}$ to its first $(k-1)$ levels serves as a good approximation to f on a sufficiently small hypercube around the origin.

Theorem 17. *Let $f \in \mathcal{F}$ that is closed under restrictions. Then for all $c \in (0, 1)$, we have*

$$\max_{\mathbf{x} \in [-c,c]^n} |f_{\geq k}(\mathbf{x})| \leq \left(\frac{c}{1-c}\right)^k M_k(\mathcal{F}).$$

In particular, it follows that

$$\varepsilon_{c,k}(\mathcal{F}) \leq \left(\frac{c}{1-c}\right)^k M_k(\mathcal{F}).$$

From Theorem 17, one immediately obtains a lower bound on $c_k(\varepsilon, \mathcal{F})$:

Corollary 18. *For any class $\mathcal{F}$ that is closed under restrictions, and any $\varepsilon > 0$ and $k \leq n$,*

$$c_k(\varepsilon, \mathcal{F}) = \Omega\left(\left(\frac{\varepsilon}{M_k(\mathcal{F})}\right)^{1/k}\right)$$

Proof. Observe that by setting $c = \Omega\left(\left(\frac{\varepsilon}{M_k(\mathcal{F})}\right)^{1/k}\right)$ in Theorem 17, the right side is bounded by ε. Because $f_{\geq k} = f - f_{<k}$ and $f_{<k}$ has degree strictly less than k, it follows immediately from the definition of $c_k(\varepsilon, \mathcal{F})$ that $c_k(\varepsilon, \mathcal{F})$ is at least c. □

We now return to the proof of Theorem 17. To prove this result, we require the following intermediate claims. The first simply shows that we may always bound the contribution of the level-k part of any function in $\mathcal{F}$ by simply rescaling the argument:

Lemma 19. *Let $f \in \mathrm{conv}(\mathcal{F})$. Then, for all $c \in (0, 1)$ and $\mathbf{x} \in [-c, c]^n$, we have*

$$|f_k(\mathbf{x})| \leq c^k M_k(\mathcal{F}).$$

Proof. Observe that $c^{-1}\mathbf{x} \in [-1, 1]^n$ by assumption, and by homogeneity of f_k as a polynomial, we have

$$|f_k(\mathbf{x})| = c^k |f_k(c^{-1}\mathbf{x})| \leq c^k M_k(\mathrm{conv}(\mathcal{F})) = c^k M_k(\mathcal{F}).$$ □

The next simple yet powerful claim shows that one can "recenter" functions in $\mathcal{F}$ and they remain in $\mathrm{conv}(\mathcal{F})$ (and therefore, enjoy the same Fourier bounds). This random restriction technique is a key tool in [CHHL19].

Lemma 20. *Let $f \in \mathrm{conv}(\mathcal{F})$, $\mathbf{a} \in [-1,1]^n$ and $\mathbf{b} \in [0,1]$ such that $|a_i| + b_i \leq 1$ for all $i \in [n]$. Define $\tilde{f}$ by $\tilde{f}(\mathbf{x}) = f(\mathbf{a} + \mathbf{b} \circ \mathbf{x})$, where $\circ$ denotes componentwise multiplication. Then, $\tilde{f} \in \mathrm{conv}(\mathcal{F})$.*

Proof. Given $\mathbf{a}$ and $\mathbf{b}$, define a distribution D_i on $Z_i = \{-1, 1, x_i\}$ where x_i is treated as formal variable, such that $\mathbb{E}_{y_i \sim D_i}[y_i] = a_i + b_i x_i$; note that this is possibly by the assumption that $|a_i| + b_i \leq 1$. Let $D = \prod_i D_i$ be the product distribution of the D_i. For any $\mathbf{z} \in \prod_i Z_i$, define $f_{\mathbf{z}}(\mathbf{x})$ as the function obtained by setting $x_i = z_i$ for each i; in particular, each variable gets set to ± 1 or remains a formal variable. By our assumption on the closure of $\mathcal{F}$, we clearly have $f_{\mathbf{z}} \in \mathcal{F}$ for any $\mathbf{z}$. By multilinearity and independence of the product distribution, we have $f(\mathbf{a} + \mathbf{b} \circ \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim D}[f_{\mathbf{z}}(\mathbf{x})]$. Thus $\tilde{f} \in \mathrm{conv}(\mathcal{F})$. □

As mentioned before, our approach will be to bound the higher-order terms of the Fourier expansion at the fractional points of the fractional PRG via the error term that arises in Taylor's theorem. Denote by $h^{(k)}$ the k-th derivative of any C^k function $h : \mathbb{R} \to \mathbb{R}$. We then have the following claim:

Lemma 21. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be multilinear and let $\mathbf{x} \in \mathbb{R}^n$. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(t) = f(t\mathbf{x})$. Then,*

$$g^{(k)}(0) = k! \cdot f_k(\mathbf{x}).$$

Proof. From the definition, it follows that

$$g(t) = \sum_{S \subseteq [n]} t^{|S|} \hat{f}(S) \mathbf{x}^S.$$

Differentiating g with respect to t, we get

$$g^{(k)}(t) = \sum_{S : |S| \geq k} \Big(\prod_{i=0}^{k-1} (|S| - i) \Big) t^{|S|-k} \hat{f}(S) \mathbf{x}^S.$$

Setting $t = 0$ eliminates all of the monomials with $|S| > k$, giving us the required bound. □

The last intermediate result we require connects the function defined in the previous part with our assumed Fourier bounds:

Lemma 22. *Let $f \in \mathrm{conv}(\mathcal{F})$, $c \in (0,1)$ and $\mathbf{x} \in [-c,c]^n$. Define g as in Theorem 21. Then,*

$$\max_{s \in [0,1]} \left| g^{(k)}(s) \right| \leq \left(\frac{c}{1-c} \right)^k \cdot k! \cdot M_k(\mathcal{F})$$

Proof. Fix $s \in [0,1]$ and let $\lambda = 1 - c \in [0,1]$. Define the auxiliary function $\tilde{f}(\mathbf{y}) = f(s\mathbf{x} + \lambda \mathbf{y})$. Writing $\mathbf{a} = s\mathbf{x}$ and $\mathbf{b} = (\lambda, \ldots, \lambda)$, we clearly have $s|x_i| + \lambda \leq 1$, so we may apply Theorem 20 to see that $\tilde{f} \in \mathrm{conv}(\mathcal{F})$. Now

writing $\tilde{g}(t) = \tilde{f}(t\mathbf{x}) = f(s\mathbf{x} + \lambda t\mathbf{x})$, we also have $\tilde{g}(t) = g(s + t\lambda)$. By the chain rule, differentiating both sides k times and then setting $t = 0$, we have

$$\lambda^k g^{(k)}(s) = \tilde{g}^{(k)}(0).$$

On the other hand, by Theorem 21, we have $\tilde{g}^{(k)}(0) = k! \cdot \tilde{f}_k(\mathbf{x})$, and as $\tilde{f} \in \mathrm{conv}(\mathcal{F})$ by Theorem 20, we conclude using Theorem 19 that

$$\left|g^{(k)}(s)\right| = \left|\frac{\tilde{g}^{(k)}(0)}{\lambda^k}\right| \leq \left(\frac{c}{1-c}\right)^k \cdot k! \cdot M_k(\mathcal{F}). \qquad \square$$

With these intermediate claims taken care of, we may now put them together to obtain Theorem 17.

Proof of Theorem 17. The second statement follows immediately from the first by setting $g = f_{<k}$ for any given f, and noticing that $f - g = f_{\geq k}$. Therefore, we focus on the first statement.

Let $f \in \mathcal{F}, \mathbf{x} \in [-c, c]^n$ and define $g(t) = f(t\mathbf{x})$. Then, by Taylor expanding g about $t = 0$ and evaluating g at $t = 1$, we have

$$g(1) = \sum_{i<k} \frac{g^{(i)}(0)}{i!} + R_k, \tag{4}$$

where R_k is the error term and is given in Lagrange form by

$$R_k = \frac{g^{(k)}(s)}{k!}$$

for some $s \in (0, 1)$. By Theorem 21, we easily see that the first term in the right hand side of (4) is precisely $f_{<k}(\mathbf{x})$, and as $g(1) = f(\mathbf{x})$, we clearly then must have $R_k = f_{\geq k}(\mathbf{x})$. Therefore, by Theorem 22, we obtain

$$|f_{\geq k}(\mathbf{x})| = \left|\frac{g^{(k)}(s)}{k!}\right| \leq \left(\frac{c}{1-c}\right)^k M_k(\mathcal{F}),$$

as desired. $\square$

3.1 Lower Bounds via Chebyshev Polynomials

In this subsection, we show that our bounds on the uniform error of any low-degree polynomial approximator are essentially tight for a reasonable range of $c < 1$. Recall that Theorem 17 shows that the low-degree Fourier expansion is an excellent approximator to the original function for c small enough; we now show that this bound cannot be significantly improved for a reasonable range of c using *any* approximator. Our main result of this section is the following converse:

Theorem 23. *Let $\mathcal{F}$ be any class of n-variate multilinear functions that are closed under restrictions. Then for any $c \leq \min\left(\frac{1}{3}, 3^{-k}\frac{M_k(\mathcal{F})}{M_{k+1}(\mathcal{F})}\right)$, we have*

$$\varepsilon_{c,k}(\mathcal{F}) \geq \left(\frac{c}{2}\right)^k M_k(\mathcal{F}).$$

Recall that on the interval $[-1,1]$, the Chebyshev polynomials give the minimum ℓ_∞ norm among all polynomials with same leading coefficient in magnitude:

Fact 24 (Theorem 1.5.4 of [RS02])**.** *If a polynomial $f : \mathbb{R} \to \mathbb{R}$ is monic of degree n, then $\max_{x\in[-1,1]} |f(x)| \geq 2^{-n+1}$, with equality if and only if $f = T_n$, the normalized n-th Chebyshev polynomial.*

Proof of Theorem 23. Let $(f, \mathbf{x})$ attain the maximum in the definition of $M_k(\mathcal{F})$, namely

$$M_k(\mathcal{F}) = \left| \sum_{S\subseteq[n]:|S|=k} \widehat{f}(S)\mathbf{x}^S \right|.$$

First, note that the claim is trivial if every function in $\mathcal{F}$ is of degree at most k, because then $f_{\geq k}$ is a homogeneous polynomial of degree k and this lower bound is trivial. Under this assumption, $M_{k+1}(\mathcal{F}) > 0$. Fix $c \in (0,1)$ and let $p : [-1,1]^n \to \mathbf{R}$ be any multilinear polynomial of degree strictly less than k. Define the univariate function $g : [-1,1] \to \mathbb{R}$ by

$$g(t) = f(tc\mathbf{x}) - p(tc\mathbf{x}).$$

By taking the Fourier expansion of f, it is easy to see that the coefficient of t^ℓ for $\ell \geq k$ is precisely

$$c^\ell \sum_{S\subseteq[n]:|S|=\ell} \widehat{f}(S)\mathbf{x}^S,$$

so that the coefficient of t^k is equal to $c^k M_k(\mathcal{F})$ in magnitude. We then have

$$\begin{aligned}
\sup_{\mathbf{z}\in[-c,c]^n} |f(\mathbf{z}) - p(\mathbf{z})| &\geq \max_{\mathbf{z}\in[-c\mathbf{x},c\mathbf{x}]} |f(\mathbf{z}) - p(\mathbf{z})| \\
&= \sup_{t\in[-1,1]} |g(t)| \\
&\geq \sup_{t\in[-1,1]} |g_{\leq k}(t)| - \sup_{t\in[-1,1]} |g_{\geq k+1}(t)|.
\end{aligned}$$

By Theorem 24, the first term is at least $c^k M_k(\mathcal{F})/2^{k-1}$. On the other hand, the second term can be bounded using Theorem 17 by

$$\sup_{t\in[-1,1]} |g_{\geq k+1}(t)| \leq \left(\frac{c}{1-c}\right)^{k+1} M_{k+1}(\mathcal{F}).$$

Therefore, we obtain

$$\sup_{\mathbf{z}\in[-c,c]^n} |f(\mathbf{z}) - p(\mathbf{z})| \geq 2\left(\frac{c}{2}\right)^k M_k(\mathcal{F}) - \left(\frac{c}{1-c}\right)^{k+1} M_{k+1}(\mathcal{F}).$$

It is straightforward to verify that for $c \leq \min\left(1/3, 3^{-k}\frac{M_k(\mathcal{F})}{M_{k+1}(\mathcal{F})}\right)$, the second term is bounded by half of the first. Because p was an arbitrary low-degree multilinear polynomial, the claim follows. □

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